

Differential Equations Lesson 16

Question 1:

Find the general solution of the linear homogeneous differential equation $2y''' - 9y'' + 7y' + 6y = 0$.

To find the solution of the linear homogeneous differential equation with constant coefficients, let $y(x) = e^{r \cdot x}$.

Hence, $y'(x) = r e^{r \cdot x}$, $y''(x) = r^2 e^{r \cdot x}$, and $y'''(x) = r^3 e^{r \cdot x}$.

Substitute $y(x)$, $y'(x)$, $y''(x)$, and $y'''(x)$ into $2y''' - 9y'' + 7y' + 6y = 0$: $2[r^3 e^{r \cdot x}] - 9r^2 e^{r \cdot x} + 7[r e^{r \cdot x}] + 6e^{r \cdot x} = 0$

$$\Rightarrow e^{r \cdot x} [2r^3 - 9r^2 + 7r + 6] = 0$$

$$\Rightarrow 2r^3 - 9r^2 + 7r + 6 = 0 \quad \text{since } e^{r \cdot x} > 0$$

$$\text{Hint: } 2r^3 - 9r^2 + 7r + 6 = (2r + 1)(r - 3)(r - 2)$$

- a) The roots of the indicial equation $2r^3 - 9r^2 + 7r + 6 = 0$ are $r_1 = \underline{\quad ? \quad}$, $r_2 = \underline{\quad ? \quad}$, and $r_3 = \underline{\quad ? \quad}$.
- b) Is $y_1(x) = e^{r_1 \cdot x}$ is solution of $2y''' - 9y'' + 7y' + 6y = 0$?
- c) Is $y_2(x) = e^{r_2 \cdot x}$ is solution of $2y''' - 9y'' + 7y' + 6y = 0$?
- d) Is $y_3(x) = e^{r_3 \cdot x}$ is solution of $2y''' - 9y'' + 7y' + 6y = 0$?
- e) Are $y_1(x) = e^{r_1 \cdot x}$, $y_2(x) = e^{r_2 \cdot x}$, and $y_3(x) = e^{r_3 \cdot x}$ linearly independent?
- f) The general solution of the differential equation $2y''' - 9y'' + 7y' + 6y = 0$ is $\underline{\quad ? \quad}$

Question 2:

Find the general solution of the linear homogeneous differential equation $y''' + y'' - 8y' - 12y = 0$.

To find the solution of the linear homogeneous differential equation with constant coefficients, let $y(x) = e^{rx}$.

Hence, $y'(x) = re^{rx}$, $y''(x) = r^2e^{rx}$, and $y'''(x) = r^3e^{rx}$.

Substitute $y(x)$, $y'(x)$, $y''(x)$, and $y'''(x)$ into $y''' + y'' - 8y' - 12y = 0$: $[r^3e^{rx}] + r^2e^{rx} - 8[re^{rx}] - 12e^{rx} = 0$

$$\Rightarrow e^{rx}[r^3 + r^2 - 8r - 12] = 0$$

$$\Rightarrow r^3 + r^2 - 8r - 12 = 0 \quad \text{since } e^{rx} > 0$$

$$\text{Hint: } r^3 + r^2 - 8r - 12 = (r + 2)(r + 2)(r - 3)$$

- a) The roots of the indicial equation $r^3 + r^2 - 8r - 12 = 0$ are $r_1 = \underline{\quad? \quad}$, $r_2 = \underline{\quad? \quad}$, and $r_3 = \underline{\quad? \quad}$.
- b) Is $y_1(x) = e^{r_1 \cdot x}$ is solution of $y''' + y'' - 8y' - 12y = 0$?
- c) Is $y_2(x) = x \cdot e^{r_1 \cdot x}$ is solution of $y''' + y'' - 8y' - 12y = 0$?
- d) Is $y_3(x) = e^{r_3 \cdot x}$ is solution of $y''' + y'' - 8y' - 12y = 0$?
- e) Are $y_1(x) = e^{r_1 \cdot x}$, $y_2(x) = x \cdot e^{r_1 \cdot x}$, and $y_3(x) = e^{r_3 \cdot x}$ linearly independent?
- f) The general solution of the differential equation $y''' + y'' - 8y' - 12y = 0$ is $\underline{\quad? \quad}$

Question 3:

Find the general solution of the linear homogeneous differential equation $y''' + 2y'' - 14y' - 40y = 0$.

To find the solution of the linear homogeneous differential equation with constant coefficients, let $y(x) = e^{rx}$.

Hence, $y'(x) = re^{rx}$, $y''(x) = r^2e^{rx}$, and $y'''(x) = r^3e^{rx}$.

Substitute $y(x)$, $y'(x)$, $y''(x)$, and $y'''(x)$ into $y''' + 2y'' - 14y' - 40y = 0$: $[r^3e^{rx}] + 2r^2e^{rx} - 14[re^{rx}] - 40e^{rx} = 0$

$$\Rightarrow e^{rx}[r^3 + 2r^2 - 14r - 40] = 0$$

$$\Rightarrow r^3 + 2r^2 - 14r - 40 = 0 \quad \text{since } e^{rx} > 0$$

$$\text{Hint: } r^3 + 2r^2 - 14r - 40 = (r^2 + 6r + 10)(r - 4)$$

a) The roots of the indicial equation $r^3 + 2r^2 - 14r - 40 = 0$ are $r_1 = \underline{\quad? \quad}$, $r_2 = \underline{\quad? \quad}$, and $r_3 = \underline{\quad? \quad}$.

Note: r_1 and r_2 will be complex numbers of the form $\alpha \pm \beta i$.

b) Is $y_1(x) = e^{\alpha x} \cdot \cos x$ is solution of $y''' + 2y'' - 14y' - 40y = 0$?

c) Is $y_2(x) = e^{\alpha x} \cdot \sin x$ is solution of $y''' + 2y'' - 14y' - 40y = 0$?

d) Is $y_3(x) = e^{r_3 x}$ is solution of $y''' + 2y'' - 14y' - 40y = 0$?

e) Are $y_1(x) = e^{r_1 x}$, $y_2(x) = x \cdot e^{r_1 x}$, and $y_3(x) = e^{r_3 x}$ linearly independent?

f) The general solution of the differential equation $y''' + 2y'' - 14y' - 40y = 0$ is $\underline{\quad? \quad}$

Question 4:

Find the general solution of the linear homogeneous differential equation $6y''' + 49y'' + 44y' - 35y = 0$.
Then find the particular solution with the initial conditions $y''(0) = 1$, $y'(0) = -1$, and $y(0) = 2$.

Hint: $6r^3 + 49r^2 + 44r - 35 = (3r + 5)(2r - 1)(r + 7)$

- a) The roots of the indicial equation $6r^3 + 49r^2 + 44r - 35 = 0$ are $r_1 = \underline{\quad? \quad}$, $r_2 = \underline{\quad? \quad}$, and $r_3 = \underline{\quad? \quad}$.
b) The general solution of the differential equation $6y''' + 49y'' + 44y' - 35y = 0$ is $\underline{\quad? \quad}$

Hint: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x) + C_3 \cdot y_3(x)$

- c) Find C_1 , C_2 , and C_3 by using the the initial conditions $y''(0) = 1$, $y'(0) = -1$, and $y(0) = 2$.

The three equations in terms of C_1 , C_2 , and C_3 are $\underline{\quad? \quad}$

- d) The augmented matrix of the three equations in part (c) is $\underline{\quad? \quad}$
e) The row-reduced echelon form of the augmented matrix in part (d) is $\underline{\quad? \quad}$
f) The particular solution with the initial conditions $y''(0) = 1$, $y'(0) = -1$, and $y(0) = 2$ is $\underline{\quad? \quad}$

Question 5:

Find the general solution of the linear homogeneous differential equation $3y'''' + 12y''' + 8y'' - 8y' - 15y = 0$.

Hint: $3r^4 + 12r^3 + 8r^2 - 8r - 15 = (3r^2 + 6r + 5)(r - 1)(r + 3)$.

a) The roots of the indicial equation $3r^4 + 12r^3 + 8r^2 - 8r - 15 = 0$ are $r_1 = \underline{\quad? \quad}$, $r_2 = \underline{\quad? \quad}$, $r_3 = \underline{\quad? \quad}$, and $r_4 = \underline{\quad? \quad}$.

Note: The roots r_1 and r_2 will be complex numbers of the form $\alpha \pm \beta i$.

b) The general solution of the differential equation $3y'''' + 12y''' + 8y'' - 8y' - 15y = 0$ is $\underline{\quad? \quad}$

Hint: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x) + C_3 \cdot y_3(x) + C_4 \cdot y_4(x)$